ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

Project Final Report

SPL-I SWE 4304 : Software Project Lab I

Project Title

AirCLI

Group 3

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Problem Statement

Modern airport systems involve complex workflows combining passenger management, security screening, financial transactions, and administrative control. However, implementing such systems in real-world environments requires extensive integration with hardware (e.g., scanners), external databases, and real-time communication systems.

The problem addressed in this project is to design a modular and secure airport simulation system that captures the essential logic of airport operations without relying on external infrastructure. The goal is to simulate how security decisions can be modeled algorithmically using contextual data (baggage contents, origin, financial behavior), mimicking real-world screening processes such as X-ray–guided inspection.

Potential Solution

AirCLI proposes a command-line based modular system where each component of airport operations are implemented as an independent yet interconnected module.
The system introduces:

- A role-based access control (RBAC) mechanism to enforce permissions
- A two-level baggage model (declared vs hidden items)
- A suspicion scoring system simulating background checks
- A time-based flight lifecycle (boarding, flying, landed)

By isolating responsibilities into modules such as banking, ticketing, baggage, and security, the system ensures scalability and maintainability. While the system operates in a simulated environment, its architecture reflects real-world extensibility where external APIs and hardware could be integrated

Flowchart

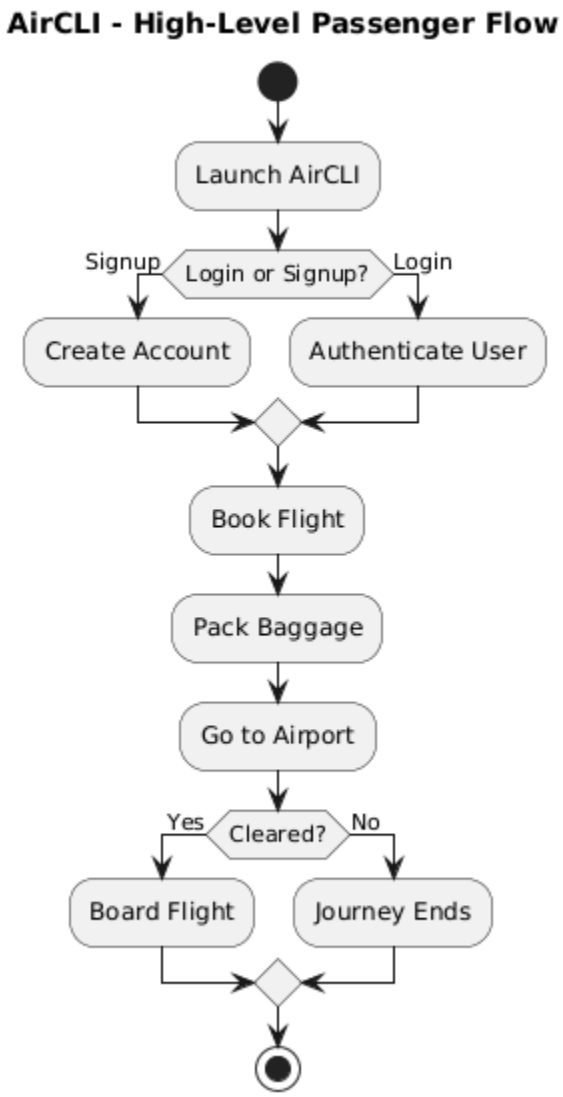

Fig: Passenger Flow

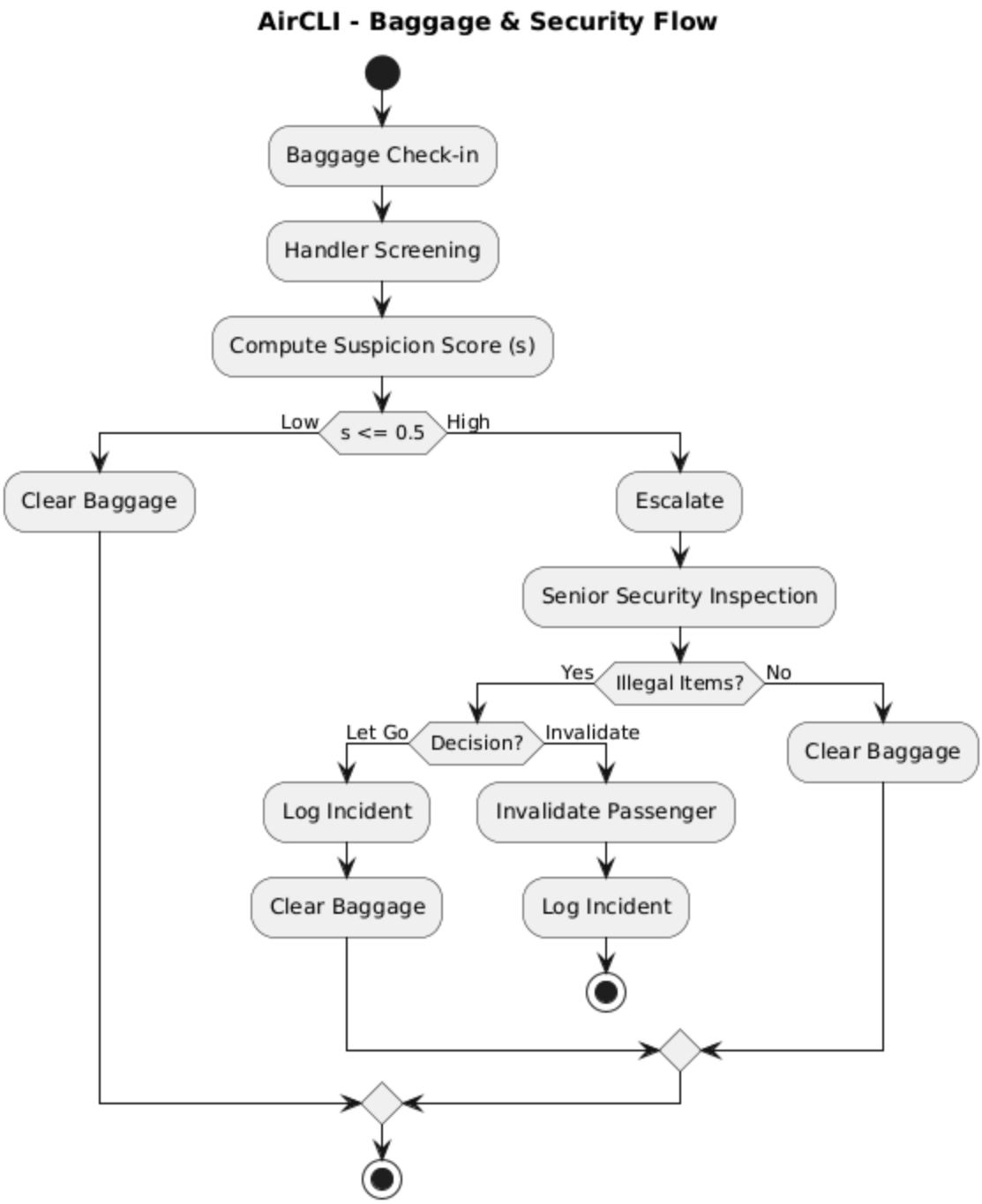

Fig: Baggage and Security Flow

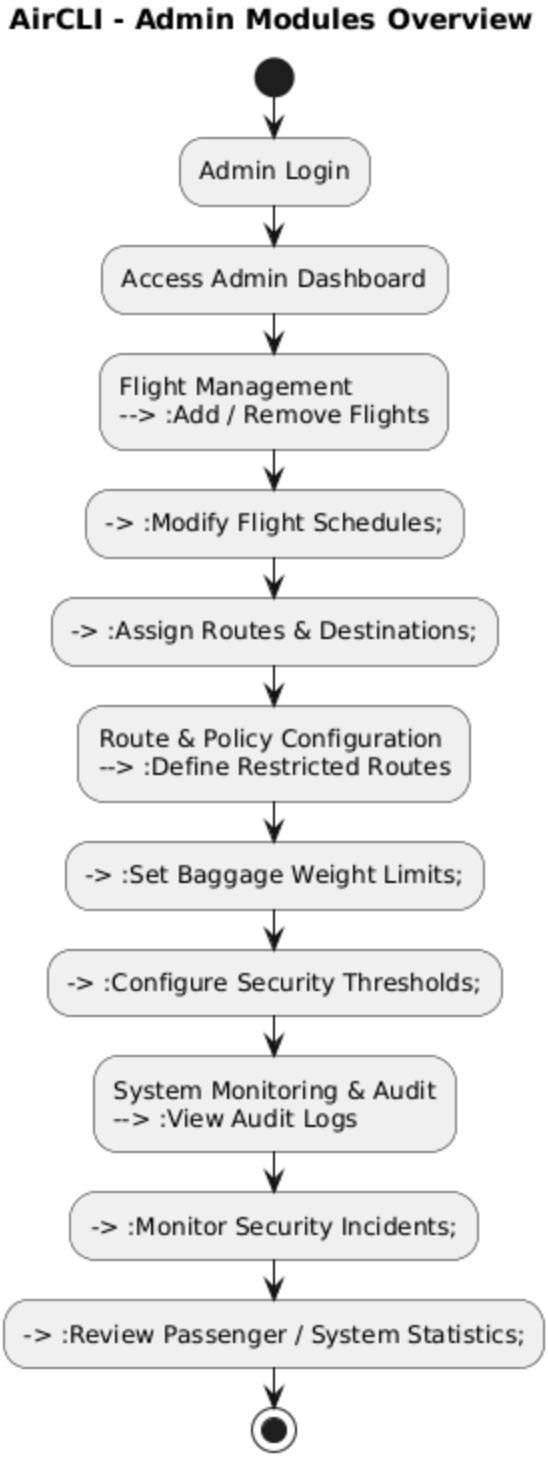

Fig: Admin Module Flow

Implemented Features

1. User Roles

Role	Description
Passenger	Books flights, packs baggage, creates bank, links bank to airCLI account, boards flights
Baggage Handler	Can check each passenger's baggage and issue a fixed fine, a permanent ban, or clear the passenger.
Admin	Manages flights, maintains airport budget, controls roles and users, starts flights.

2. Core Modules

2.1. User-Side Modules (Passenger-facing)

These modules represent all functionalities directly accessible to passengers through the CLI.

2.1.1. User Account Module (airCLI)

The User Account Module manages passenger identity within the system. It supports user registration and authentication, maintains user credentials, and associates each passenger with their booking history, baggage records, and linked banking information. This module serves as the entry point for all passenger-side interactions.

- User registration (signup)
- Authentication (login)
- Profile Viewing
- Role Assignment (passenger)

2.1.2. Banking and Payment Module (Bank CLI)

The Banking and Payment Module simulates financial transactions required for flight bookings. It validates payment credentials, verifies account balances, processes debits and credits, and records transaction history. The module ensures that bookings proceed only after successful and traceable payment confirmation.

- Simulated Bank Account Linkage
- PIN Validation
- Balance Management
- Payment Processing for Flight Bookings
- Transaction Logging

2.1.3. Flight Search & Booking Module (airCLI)

This module enables passengers to discover and reserve flights based on travel criteria such as origin, destination, and date. It handles flight selection, coordinates with the payment system, and initiates ticket creation upon successful booking. The module forms the core of the passenger travel planning process.

- Search flights by: Origin, Destination, Date
- Flight Selection
- Payment Initiation
- Ticket Issuance after Successful Payment

2.1.4. Ticket Management Module (airCLI)

The Ticket Management Module is responsible for generating, storing, and maintaining ticket records. It tracks ticket states such as validity, expiration, early arrival, and invalidation due to security actions. This module ensures that each ticket accurately reflects the passenger's authorization to travel.

- Ticket Generation
- Ticket Storage
- Ticket Status Tracking (based on flight starting, landing, and security clearance)
- Ticket-to-User Mapping

2.1.5. Baggage Packing Module (baggageCLI)

The Baggage Packing Module simulates the process of preparing luggage prior to airport arrival. It allows passengers to organize baggage contents using a two-level structure, separating declared items from

private items. The module calculates baggage weight and prepares baggage data for subsequent airport-side inspection.

- Baggage creation
- Two-level item loading: Public, Private
- Weight Calculation based on a pre-existing list of items.
- Final Baggage Submission

2.1.6. Passenger Activity Module

This module provides optional informational and navigational features for passengers. It allows users to view upcoming flights, review past bookings, monitor baggage status, and transition into the airport phase of the system. The module enhances usability without influencing core security decisions.

- View Upcoming Flights
- View Travel History
- Board Flights

2.2. Security-Side Modules (Airport Authority)

2.2.1. Ticket Checking Module

The Ticket Checking Module operates at the airport entry point to verify passenger authorization. It validates ticket authenticity and checks time-based constraints such as early arrival or expiration. Based on the evaluation, the module either permits entry or denies access.

- Ticket Authentication
- Time-based Validation
- Entry Authorization or Rejection

2.2.2. Baggage Screening Module (Handler)

The Baggage Screening Module conducts inspection of passenger luggage. It evaluates factors such as weight irregularities, route-item inconsistencies, and passenger security history to compute a suspicion score. This module determines whether baggage is cleared or escalated for deeper inspection.

- Baggage inspection
- Weight Mismatch Detection
- Route-item consistency Evaluation

- Passenger History Lookup
- Suspicion Score Calculation
- Escalation Decision
- Issuance of Fine, Permanent Ban (only reversible by Admin), or Clear Passenger
- Incident Logging

2.3. Admin-Side Modules (System Control)

2.3.1. Flight Management Module

This module enables administrators to manage flight-related data within the system. It supports the creation, modification, and removal of flights, as well as schedule and destination management. The module ensures that accurate and up-to-date flight information is available for booking and validation.

- Add/ Remove Flights
- Modify Flight Schedules
- Assign Route/ Destination

2.3.2. System Monitoring Module

This module provides administrative visibility into system operations. It allows administrators to review logs, monitor security incidents, and observe overall system activity. The module supports oversight and accountability without modifying operational data.

- View Audit Logs
- Monitor Security Incidents
- Review Passenger Statistics

2.4. General Modules (Non-Role-Specific)

2.4.1. Data Storage Module

The Data Storage Module manages persistent storage using structured text files. It handles reading, writing, and updating system data including users, flights, tickets, baggage records, and logs. The module ensures data consistency without reliance on external databases.

- XML-based persistence
- Structured File Formats

- Data loading and saving
- Backup Handling

2.4.2. Access Control Module

The Access Control Module enforces role-based permissions across the system. It ensures that users, airport staff, and administrators can only access functionality appropriate to their assigned roles. This module prevents unauthorized actions and protects sensitive system data.

- Role Validation
- Permission Enforcement
- Restricted data access

2.4.3. Audit and Logging Module

This module records all critical system actions in append-only logs. It tracks financial transactions, baggage inspections, and security decisions to ensure traceability and accountability. Logged data supports review, debugging, and compliance verification.

- Payment Logs
- Baggage Inspection Logs
- Security Incident Logs

2.4.4. Utility and Validation Module

The Utility and Validation Module provides shared support functions across the system. It handles input validation, date and time checks, identifier generation, and consistency verification. This module improves reliability and reduces duplication of common logic.

- Input Validation
- Date & Time Validation
- ID Generation
- Consistency Checks

Tools & Technology

1. Programming Language: C++ (C++17)
2. Compiler: GNU G++
3. IDE: Visual Studio Code
4. Version Control: Git
5. Scripting: Bash (build.sh)
6. Data Storage: XML-based persistence
7. Libraries: Standard Template Library (STL)
8. Diagram Tools: PlantUML

Project Timeline

Phase	Subtask	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14
Planning Design	Requirement Analysis														
	Project Presentation														
	System Design & Architecture														
	UML & Flow Diagrams														
Core User Modules	User Account Module														
	Banking & Payment Module														
	Flight Search & Booking														
	Ticket Management														

Phase	Subtask	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14
Baggage & Security	Baggage Packing Module														
	Baggage Screening Module														
	Senior Security Module														
Airport Operations	Ticket Checking Module														
	Passenger Activity Module														
Admin /System Module	Flight Management Module														
	Route & Policy Configuration														
	Access Control & Logging														
Testing & Documentation	Integration Testing														
	Documentation														

Contribution

• Ittesaf Yeasir Ithun:

- User Account Module
- Flight Search & Booking Module
- Flight Management Module
- Ticket Management Module
- System Monitoring Module
- Access Control Module
- Utility and Validation Module

• Nafian Mahid:

- Banking & Payment Module
- Banking Data Persistence
- Payment Portal Integration in airCLI
- Saved Payments Feature
- Logging Module for Bank Users

• Md. Faiaj Jarif:

- Baggage Packing Module
- Baggage Screening Module
- Data Storage Module
- Audit and Logging Module
-

Future Work

While AirCLI successfully simulates airport operations, several enhancements can improve realism and scalability:

• Integration with real databases (e.g., PostgreSQL)
• GUI-based interface for improved usability
• API integration for real-time flight data
• Integration with external hardware simulations (e.g., scanners)
• Networked multi-user environment

Source Code Repository

https://github.com/ithun-y-ittesaf/airCLI